

# WASP-12b

R-0321

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_231230 · created 2026-07-18T01:59:17+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460308.7796 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.155 +/- 0.045                 |
| Transit depth (Rp/Rs)^2                           | 2.4 +/- 1.38 [%]                |
| Orbital Inclination (inc)                         | 85.94 +/- 2.83                  |
| Airmass coefficient 1 (a1)                        | 1.193 +/- 0.083                 |
| Airmass coefficient 2 (a2)                        | -0.2 +/- 0.24                   |
| Scatter in the residuals of the lightcurve fit is | 7.08 %                          |
| Best Comparison Star                              | #3 - [272, 295]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.1323 +/- 0.0066               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R* fit vs published    | 0.155 ± 0.045 vs 0.1178 (deviation 0.68σ) |
| Duration fit vs published | 0.1323 d vs 0.125 d (deviation 0.91σ)     |
| Mid-time O–C              | -0.2 min                                  |
| Depth SNR                 | 2.8                                       |
| Residual scatter          | 7.08 %                                    |

### Light curve

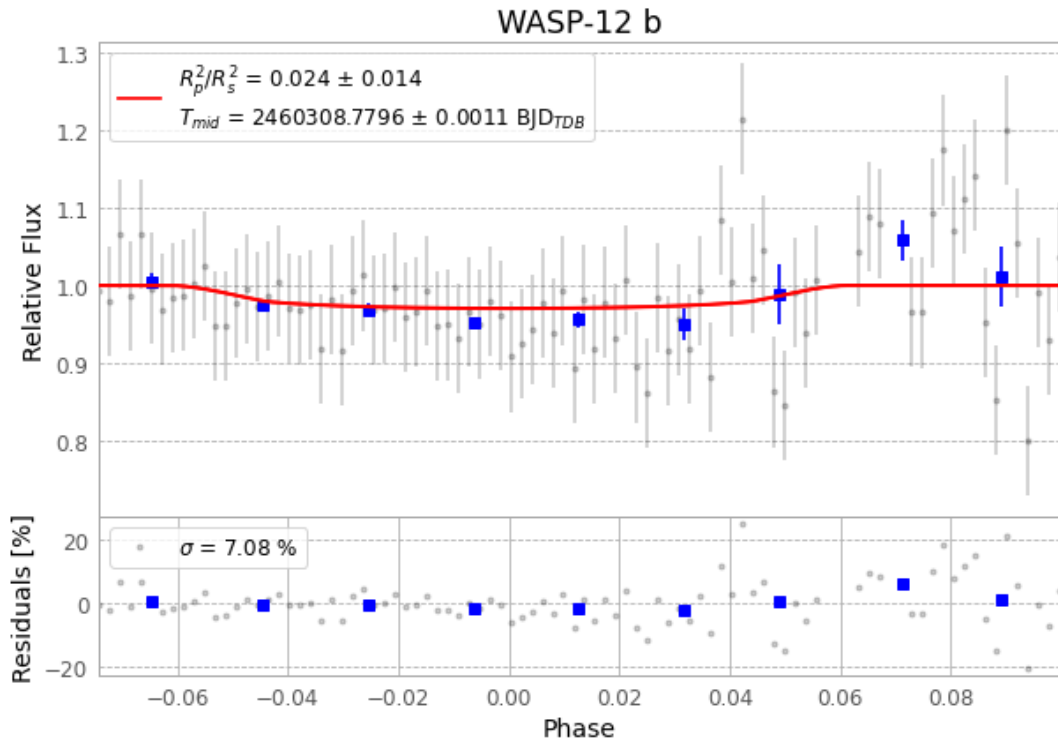


Figure 1 — FinalLightCurve\_WASP-12 b\_2023-12-29.png (SHA-256 aca19df3a084be3e...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 257], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b3601b0825cff606...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1bbef5c

## Data provenance

92 FITS frames (60 MB), WASP-12231230043616.FITS → WASP-12231230090915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231230.log (8611d4de7064...), FinalLightCurve\_WASP-12 b\_2023-12-29.png (aca19df3a084...), FinalLightCurve\_WASP-12 b\_2023-12-29.csv (61c61f925ecf...)

## Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 01:59Z.